PROBLEMS SHEET 3 • OFFICIAL SOLUTIONS

Department of Mathematics • UC3M Aerospace Engineering

# Chapter 3 Problem Set: Second-Order Linear ODEs & Oscillations

Comprehensive, mathematically rigorous analytical solutions to all 11 problems from `ProblemsCh3.pdf`. Covering fundamental sets of solutions, Wronskian determinants, Abel's identity, Peano's counterexample, 15 complete initial value problems, saddle point invariant manifolds, polar amplitude-phase transformations, conservative simple pendulums, mechanical damping regimes, falling chain dynamics, linear drag terminal velocity, and Euler-Cauchy differential equations.

| | |
|---|---|
| Source Document: | sources/cuatrimestre-1/04-advanced-maths/problemas/ProblemsCh3.pdf (probls_ch3_2627.pdf) |

OFFICIAL MATERIAL ProblemsCh3.pdf Robinson (2004).pdf

EXERCISE 1 — Wronskian Determinant • Fundamental Bases

## Linear Independence of Canonical Solution Pairs via Wronskian

Original statement · ProblemsCh3.pdf

---

PHASE 1: STATEMENT & SETUP

> By finding the Wronskian of the following pairs of functions, show that they are linearly independent:
>
> 1. $x_1(t) = e^{k_1 t}$ and $x_2(t) = e^{k_2 t}$ with $k_1 \neq k_2$
> 2. $x_1(t) = e^{kt}$ and $x_2(t) = te^{kt}$
> 3. $x_1(t) = e^{\rho t}\sin(\omega t)$ and $x_2(t) = e^{\rho t}\cos(\omega t)$ with $\omega \neq 0$

PHASE 2: FUNDAMENTAL FORMULATION

For two continuously differentiable functions $x_1, x_2 \in C^1(I)$, the Wronskian determinant is defined as:

$$W[x_1, x_2](t) = \det \begin{pmatrix} x_1(t) & x_2(t) \\ \dot{x}_1(t) & \dot{x}_2(t) \end{pmatrix} = x_1(t)\dot{x}_2(t) - x_2(t)\dot{x}_1(t)$$

If $W[x_1, x_2](t_0) \neq 0$ at even a single point $t_0 \in I$, the functions $x_1(t)$ and $x_2(t)$ are strictly **linearly independent** on the interval $I$.

PHASE 3: STEP-BY-STEP ANALYTICAL DERIVATION

**Part (i): Distinct Exponential Modes** $x_1(t) = e^{k_1 t},\ x_2(t) = e^{k_2 t}$ **(** $k_1 \neq k_2$ **)**

Differentiating each function:

$$\dot{x}_1(t) = k_1 e^{k_1 t}, \qquad \dot{x}_2(t) = k_2 e^{k_2 t}$$

Formulating the Wronskian determinant:

$$W[x_1, x_2](t) = e^{k_1 t}\left(k_2 e^{k_2 t}\right) - e^{k_2 t}\left(k_1 e^{k_1 t}\right) = (k_2 - k_1)e^{(k_1+k_2)t}$$

Since the exponential function satisfies $e^{(k_1+k_2)t} > 0$ for all $t \in \mathbb{R}$, and $k_1 \neq k_2 \implies k_2 - k_1 \neq 0$:

$$W[x_1, x_2](t) \neq 0 \quad \forall t \in \mathbb{R} \implies \text{Linearly Independent on } \mathbb{R}$$

**Part (ii): Repeated Root Secular Mode** $x_1(t) = e^{kt},\ x_2(t) = te^{kt}$

Differentiating using the product rule on $x_2(t)$:

$$\dot{x}_1(t) = ke^{kt}, \qquad \dot{x}_2(t) = 1 \cdot e^{kt} + t(ke^{kt}) = (1 + kt)e^{kt}$$

Formulating the Wronskian determinant:

$$W[x_1, x_2](t) = e^{kt}\left[(1 + kt)e^{kt}\right] - te^{kt}\left[ke^{kt}\right] = e^{2kt}\left[(1 + kt) - kt\right] = e^{2kt}$$

Since $e^{2kt} > 0$ for every real $t$, $W(t)$ is strictly positive everywhere:

$$W[x_1, x_2](t) = e^{2kt} \neq 0 \quad \forall t \in \mathbb{R} \implies \text{Linearly Independent on } \mathbb{R}$$

**Part (iii): Conjugate Complex Modes** $x_1(t) = e^{\rho t}\sin(\omega t),\ x_2(t) = e^{\rho t}\cos(\omega t)$ $(\omega \neq 0)$

Differentiating via product rule:

$$\dot{x}_1(t) = e^{\rho t}\left[\rho\sin(\omega t) + \omega\cos(\omega t)\right]$$

$$\dot{x}_2(t) = e^{\rho t}\left[\rho\cos(\omega t) - \omega\sin(\omega t)\right]$$

Assembling $W(t) = x_1\dot{x}_2 - x_2\dot{x}_1$ and factoring $e^{2\rho t}$:

$$W(t) = e^{2\rho t}\left[\sin(\omega t)\left(\rho\cos(\omega t) - \omega\sin(\omega t)\right) - \cos(\omega t)\left(\rho\sin(\omega t) + \omega\cos(\omega t)\right)\right]$$

$$W(t) = e^{2\rho t}\left[\rho\sin\omega t\cos\omega t - \omega\sin^2\omega t - \rho\cos\omega t\sin\omega t - \omega\cos^2\omega t\right]$$

The cross terms $\rho\sin\omega t\cos\omega t$ cancel identically. Applying $\sin^2\omega t + \cos^2\omega t = 1$:

$$W[x_1, x_2](t) = e^{2\rho t}\left[-\omega(\sin^2\omega t + \cos^2\omega t)\right] = -\omega e^{2\rho t}$$

Because $\omega \neq 0$ and $e^{2\rho t} > 0$:

$$W[x_1, x_2](t) = -\omega e^{2\rho t} \neq 0 \quad \forall t \in \mathbb{R} \implies \text{Linearly Independent on } \mathbb{R}$$

PHASE 4: INTERPRETATION & MASTER RESULTS

MASTER WRONSKIAN SUMMARY

**(i) Distinct Exponentials:** $W[e^{k_1 t}, e^{k_2 t}](t) = (\mathbf{k_2 - k_1})\mathbf{e^{(k_1+k_2)t}} \neq 0$

**(ii) Critical Damping Secular Mode:** $W[e^{kt}, te^{kt}](t) = \mathbf{e^{2kt}} > 0$

**(iii) Damped Harmonic Oscillations:** $W[e^{\rho t}\sin\omega t, e^{\rho t}\cos\omega t](t) = -\omega\mathbf{e^{2\rho t}} \neq 0$

**AEROSPACE ENGINEERING CONTEXT**

These three fundamental pairs represent the complete basis spanning the null space of any linear time-invariant (LTI) mechanical system $m\ddot{x} + c\dot{x} + kx = 0$: overdamped aerodynamic divergence/decay, critically damped actuation, and flutter oscillation.

# Abel's Identity and the Wronskian Dichotomy

Original statement · ProblemsCh3.pdf

---

## PHASE 1: STATEMENT & SETUP

Show that the Wronskian for two solutions $x_1(t)$ and $x_2(t)$ of the second order differential equation

$$\frac{d^2x}{dt^2} + p_1(t)\frac{dx}{dt} + p_2(t)x = 0$$

satisfies $\frac{dW(t)}{dt} = -p_1(t)W(t)$. Deduce either that $W(t) = 0$ for all $t$, or that $W(t) \neq 0$ for all $t$.

## PHASE 2: FUNDAMENTAL FORMULATION

Assume $p_1(t), p_2(t)$ are continuous functions on an interval $I \subseteq \mathbb{R}$. Let $x_1(t), x_2(t) \in C^2(I)$ be solutions of $L[x] = \ddot{x} + p_1\dot{x} + p_2x = 0$. We compute the time derivative of $W(t) = x_1\dot{x}_2 - x_2\dot{x}_1$ and replace the second derivatives using the governing ODE.

## PHASE 3: STEP-BY-STEP ANALYTICAL DERIVATION

**Step 1: Differentiating $W(t)$**

$$\frac{dW}{dt} = \frac{d}{dt}\left(x_1\dot{x}_2 - x_2\dot{x}_1\right) = \left(\dot{x}_1\dot{x}_2 + x_1\ddot{x}_2\right) - \left(\dot{x}_2\dot{x}_1 + x_2\ddot{x}_1\right)$$

The product terms $\dot{x}_1\dot{x}_2$ and $\dot{x}_2\dot{x}_1$ cancel:

$$\frac{dW}{dt} = x_1\ddot{x}_2 - x_2\ddot{x}_1 \tag{1}$$

**Step 2: Substituting Second Derivatives from the ODE**

$$\ddot{x}_1 = -p_1(t)\dot{x}_1 - p_2(t)x_1, \qquad \ddot{x}_2 = -p_1(t)\dot{x}_2 - p_2(t)x_2$$

Substituting into (1):

$$\frac{dW}{dt} = x_1\left(-p_1\dot{x}_2 - p_2x_2\right) - x_2\left(-p_1\dot{x}_1 - p_2x_1\right)$$

$$\frac{dW}{dt} = -p_1(t)\left[x_1\dot{x}_2 - x_2\dot{x}_1\right] - p_2(t)\left[x_1x_2 - x_2x_1\right]$$

Since $x_1x_2 - x_2x_1 \equiv 0$ and $x_1\dot{x}_2 - x_2\dot{x}_1 = W(t)$:

$$\boxed{\mathbf{\frac{dW(t)}{dt} = -p_1(t)W(t)}} \qquad (2)$$

**Step 3: Integrating to Abel's Formula**

Equation (2) is a first-order linear ODE. We do not divide by $W$ (it may vanish, and the dichotomy is deduced afterwards), so we use the integrating factor $\mu(t) = \exp\left(\int_{t_0}^{t} p_1(s)\,ds\right)$. By the Fundamental Theorem of Calculus and the chain rule $\frac{d}{dt}e^u = e^u u'$ with $u(t) = \int_{t_0}^{t} p_1(s)\,ds$:

$$\mu'(t) = p_1(t)\mu(t), \qquad \mu(t_0) = 1, \qquad \mu(t) > 0$$

Using (2) in the product rule:

$$\frac{d}{dt}\left[\mu W\right] = \mu'W + \mu W' = p_1\mu W - p_1\mu W = 0$$

Integrating from $t_0$ to $t$ (Barrow's rule): $\mu(t)W(t) - \mu(t_0)W(t_0) = \int_{t_0}^{t} 0\,ds = 0$, hence $\mu(t)W(t) = W(t_0)$ and, dividing by $\mu(t) > 0$, for every value of $W(t_0)$ (including 0):

$$\boxed{\mathbf{W(t) = \frac{W(t_0)}{\mu(t)} = W(t_0)\exp\left(-\int_{t_0}^{t} p_1(s)\,ds\right)}} \qquad (3)$$

**Step 4: Deduction of the Wronskian Dichotomy**

Because $p_1(s)$ is continuous on $I$, the integral $\int_{t_0}^{t} p_1(s)\,ds$ is finite for all $t \in I$. Consequently, the exponential term is strictly positive:

$$\exp\left(-\int_{t_0}^{t} p_1(s)\,ds\right) > 0 \quad \forall t \in I$$

- **If $W(t_0) = 0$:** $W(t) = 0 \times \exp(\dots) = 0$ for all $t \in I$.
- **If $W(t_0) \neq 0$:** Since neither factor is zero, $W(t) \neq 0$ for all $t \in I$.

PHASE 4: INTERPRETATION & MASTER RESULTS

### ABEL'S IDENTITY & WRONSKIAN DICHOTOMY

$$\boxed{\frac{dW}{dt} = -p_1(t)W(t) \implies W(t) = W(t_0)\exp\left(-\int_{t_0}^{t} p_1(s)\,ds\right)}$$

**Conclusion:** For any two solutions of a linear homogeneous second-order ODE with continuous coefficients, the Wronskian is *either identically zero everywhere on $I$, or non-zero at every point of $I$*. It can never cross zero isolatedly.

EXERCISE 3

Peano Counterexample • Rigorous Analysis

# Peano's Counterexample: Wronskian Vanishing vs Linear Independence

Original statement · ProblemsCh3.pdf

PHASE 1: STATEMENT & SETUP

We have seen that, if $x_1(t)$ and $x_2(t)$ are two solutions of a linear homogeneous second order differential equation, then they are linearly independent if and only if their Wronskian is non-zero. This is not true for general functions that are not the solutions of some differential equation.

1. Check that, if $f(t) = t^2|t|$, then $\frac{df}{dt} = 3t|t|$ (straightforward for $t \neq 0$; use the formal definition of the derivative at $t = 0$).
2. Considering $f_1(t) = t^2|t|$ and $f_2(t) = t^3$, show that, although these two functions are linearly independent on $\mathbb{R}$, their Wronskian is identically zero.

PHASE 2: FUNDAMENTAL FORMULATION

Linear independence on a function space requires that $c_1 f_1(t) + c_2 f_2(t) = 0$ for all $t \in \mathbb{R}$ implies $c_1 = c_2 = 0$. The Wronskian is $W[f_1, f_2](t) = f_1(t) f_2'(t) - f_2(t) f_1'(t)$. This problem demonstrates why the condition "$W(t) \equiv 0 \implies$ linearly dependent" fails for arbitrary functions that do not satisfy a common regular linear ODE.

PHASE 3: STEP-BY-STEP ANALYTICAL DERIVATION

**Part (i): Differentiating $f(t) = t^2|t|$**

**For $t > 0$:** $|t| = t \implies f(t) = t^3 \implies f'(t) = 3t^2 = 3t(t) = 3t|t|$.

**For $t < 0$:** $|t| = -t \implies f(t) = -t^3 \implies f'(t) = -3t^2 = 3t(-t) = 3t|t|$.

**At $t = 0$ (Formal Newton Difference Quotient):**

$$f'(0) = \lim_{h \to 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \to 0} \frac{h^2|h| - 0}{h} = \lim_{h \to 0} h|h|$$

Testing left and right limits: $\lim_{h \to 0^+} h^2 = 0$, $\lim_{h \to 0^-} (-h^2) = 0$. Both equal $0$, so $f'(0) = 0$.

Evaluating $3t|t|$ at $t = 0$: $3(0)|0| = 0$. Thus, everywhere on $\mathbb{R}$:

$$\frac{\mathbf{df}}{\mathbf{dt}} = \mathbf{3t|t|}$$

**Part (ii): Linear Independence of $f_1(t) = t^2|t|$ and $f_2(t) = t^3$ on $\mathbb{R}$**

Assume constants $c_1, c_2 \in \mathbb{R}$ such that $c_1 f_1(t) + c_2 f_2(t) = 0$ for all $t \in \mathbb{R}$:

$$c_1 t^2|t| + c_2 t^3 = 0 \quad \forall t \in \mathbb{R}$$

Pick $t = 1$: $c_1(1)^2|1| + c_2(1)^3 = c_1 + c_2 = 0 \implies c_2 = -c_1$.

Pick $t = -1$: $c_1(-1)^2|-1| + c_2(-1)^3 = c_1(1) + c_2(-1) = c_1 - c_2 = 0 \implies c_1 - (-c_1) = 2c_1 = 0 \implies c_1 = 0,\ c_2 = 0$.

Since only the trivial combination works: $f_1$ **and** $f_2$ **are linearly independent on** $\mathbb{R}$**.**

**Wronskian Determinant on $\mathbb{R}$:**

$$W[f_1, f_2](t) = f_1(t)f_2'(t) - f_2(t)f_1'(t) = \left(t^2|t|\right)\left(3t^2\right) - \left(t^3\right)\left(3t|t|\right)$$

$$W[f_1, f_2](t) = 3t^4|t| - 3t^4|t| \equiv 0 \quad \forall t \in \mathbb{R}$$

PHASE 4: INTERPRETATION & EPISTEMOLOGICAL INSIGHT

**WHY PEANO'S COUNTEREXAMPLE DOES NOT CONTRADICT ABEL'S THEOREM**

If $f_1$ and $f_2$ were solutions of a regular second-order ODE $x'' + p_1(t)x' + p_2(t)x = 0, W \equiv 0$ would force linear dependence. However, constructing the ODE satisfied by $f_1$ and $f_2$ on $t > 0$ gives $t^2x'' - 6x = 0 \iff x'' - \frac{6}{t^2}x = 0$. The coefficient $p_2(t) = -6/t^2$ is **singular at** $t = 0$. Because the coefficient blows up at $t = 0$, the ODE is not regular, Picard-Lindelöf existence-uniqueness breaks down, and Abel's theorem does not apply across $t = 0$.

EXERCISE 4

Characteristic Polynomial • 15 IVPs Solved

# 15 Second-Order Homogeneous Linear ODEs with Initial Conditions

Original statement · ProblemsCh3.pdf

PHASE 1: STATEMENT & SETUP

Find the general solution of the following differential equations, and then the unique solution satisfying the specified initial conditions:

(i) $\ddot{x} - 3\dot{x} + 2x = 0,\ x(0) = 2, \dot{x}(0) = 6$

(ii) $y'' - 4y' + 4y = 0,\ y(0) = 0, y'(0) = 3$

(iii) $z'' - 4z' + 13z = 0,\ z(0) = 7, z'(0) = 42$

(iv) $\ddot{y} + \dot{y} - 6y = 0,\ y(0) = -1, y'(0) = 8$

(v) $\ddot{y} - 4\dot{y} = 0,\ y(0) = 13, \dot{y}(0) = 0$

(vi) $\ddot{\theta} + 4\theta = 0,\ \theta(0) = 0, \dot{\theta}(0) = 10$

(vii) $\ddot{y} + 2\dot{y} + 10y = 0,\ y(0) = 3, \dot{y}(0) = 0$

(viii) $2\ddot{z} + 7\dot{z} - 4z = 0,\ z(0) = 0, \dot{z}(0) = 9$

(ix) $\ddot{y} + 2\dot{y} + y = 0,\ y(0) = 0, \dot{y}(0) = -1$

(x) $\ddot{x} + 6\dot{x} + 10x = 0,\ x(0) = 3, \dot{x}(0) = 1$

(xi) $4\ddot{x} - 20\dot{x} + 21x = 0,\ x(0) = -4, \dot{x}(0) = -12$

(xii) $\ddot{y} + \dot{y} - 2y = 0,\ y(0) = 4, \dot{y}(0) = -4$

(xiii) $\ddot{y} - 4y = 0,\ y(0) = 10, \dot{y}(0) = 0$

(xiv) $y'' + 4y' + 4y = 0,\ y(0) = 27, y'(0) = -54$

(xv) $\ddot{y} + \omega^2 y = 0,\ y(0) = 0, \dot{y}(0) = 1$

PHASE 2: FUNDAMENTAL FORMULATION

For $a\ddot{u} + b\dot{u} + cu = 0$, the exponential ansatz $u(t) = e^{rt}$ yields the auxiliary equation:

$$ar^2 + br + c = 0, \qquad \Delta = b^2 - 4ac$$

- $\Delta > 0$: $u(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}$ (Distinct Real).
- $\Delta = 0$: $u(t) = (c_1 + c_2 t)e^{rt}$ (Repeated Real).
- $\Delta < 0$: $u(t) = e^{\rho t}(c_1 \cos \omega t + c_2 \sin \omega t)$ (Complex Conjugate $r = \rho \pm i\omega$).

Constants $c_1, c_2$ are uniquely determined by Cauchy conditions $u(0) = u_0, \dot{u}(0) = v_0$.

PHASE 3: STEP-BY-STEP ANALYTICAL DERIVATION (ALL 15 ITEMS)

**(i)** $\ddot{x} - 3\dot{x} + 2x = 0$ **with** $x(0) = 2,\ \dot{x}(0) = 6$

Auxiliary eq: $r^2 - 3r + 2 = 0 \implies (r-1)(r-2) = 0 \implies r_1 = 1, r_2 = 2.$

General solution: $x(t) = c_1 e^t + c_2 e^{2t} \implies \dot{x}(t) = c_1 e^t + 2c_2 e^{2t}.$

Initial conditions: $c_1 + c_2 = 2, c_1 + 2c_2 = 6 \implies c_2 = 4,\ c_1 = -2.$

**Solution:** $\mathbf{x(t) = -2e^t + 4e^{2t}}$

**(ii)** $y'' - 4y' + 4y = 0$ **with** $y(0) = 0,\ y'(0) = 3$

Auxiliary eq: $r^2 - 4r + 4 = 0 \implies (r-2)^2 = 0 \implies r = 2$ (mult. 2).

General solution: $y(t) = (c_1 + c_2 t)e^{2t} \implies y'(t) = (2c_1 + c_2 + 2c_2 t)e^{2t}.$

Initial conditions: $y(0) = c_1 = 0; y'(0) = 2(0) + c_2 = 3 \implies c_2 = 3.$

**Solution:** $\mathbf{y(t) = 3te^{2t}}$

**(iii)** $z'' - 4z' + 13z = 0$ **with** $z(0) = 7,\ z'(0) = 42$

Auxiliary eq: $r^2 - 4r + 13 = 0 \implies (r-2)^2 + 9 = 0 \implies r = 2 \pm 3i$ ($\rho = 2, \omega = 3$).

General solution: $z(t) = e^{2t}(c_1 \cos 3t + c_2 \sin 3t).$

Derivative: $z'(t) = e^{2t}\left[(2c_1 + 3c_2)\cos 3t + (2c_2 - 3c_1)\sin 3t\right].$

Initial conditions: $z(0) = c_1 = 7; z'(0) = 2(7) + 3c_2 = 42 \implies 3c_2 = 28 \implies c_2 = \frac{28}{3}.$

**Solution:** $\mathbf{z(t) = e^{2t}\left(7\cos 3t + \frac{28}{3}\sin 3t\right)}$

**(iv)** $\ddot{y} + \dot{y} - 6y = 0$ **with** $y(0) = -1,\ y'(0) = 8$

Auxiliary eq: $r^2 + r - 6 = 0 \implies (r+3)(r-2) = 0 \implies r_1 = 2, r_2 = -3.$

General solution: $y(t) = c_1 e^{2t} + c_2 e^{-3t} \implies \dot{y}(t) = 2c_1 e^{2t} - 3c_2 e^{-3t}.$

Initial conditions: $c_1 + c_2 = -1, 2c_1 - 3c_2 = 8 \implies 2c_1 - 3(-1 - c_1) = 5c_1 + 3 = 8 \implies c_1 = 1,\ c_2 = -2.$

**Solution:** $\mathbf{y(t) = e^{2t} - 2e^{-3t}}$

**(v)** $\ddot{y} - 4\dot{y} = 0$ **with** $y(0) = 13,\ \dot{y}(0) = 0$

Auxiliary eq: $r^2 - 4r = 0 \implies r(r-4) = 0 \implies r_1 = 0, r_2 = 4.$

General solution: $y(t) = c_1 + c_2 e^{4t} \implies \dot{y}(t) = 4c_2 e^{4t}.$

Initial conditions: $\dot{y}(0) = 4c_2 = 0 \implies c_2 = 0; y(0) = c_1 = 13.$

**Solution:** $\mathbf{y(t) = 13}$ (Steady constant equilibrium)

**(vi)** $\ddot{\theta} + 4\theta = 0$ **with** $\theta(0) = 0,\ \dot{\theta}(0) = 10$

Auxiliary eq: $r^2 + 4 = 0 \implies r = \pm 2i$ ($\rho = 0, \omega = 2$).

General solution: $\theta(t) = c_1 \cos 2t + c_2 \sin 2t \implies \dot{\theta}(t) = -2c_1 \sin 2t + 2c_2 \cos 2t.$

Initial conditions: $\theta(0) = c_1 = 0; \dot{\theta}(0) = 2c_2 = 10 \implies c_2 = 5.$

**Solution:** $\theta(\mathbf{t}) = \mathbf{5}\sin\mathbf{2t}$

**(vii)** $\ddot{y} + 2\dot{y} + 10y = 0$ **with** $y(0) = 3,\ \dot{y}(0) = 0$

Auxiliary eq: $r^2 + 2r + 10 = 0 \implies (r+1)^2 + 9 = 0 \implies r = -1 \pm 3i\ (\rho = -1, \omega = 3).$

General solution: $y(t) = e^{-t}(c_1\cos 3t + c_2 \sin 3t).$

Derivative: $\dot{y}(t) = e^{-t}\left[(3c_2 - c_1)\cos 3t - (3c_1 + c_2)\sin 3t\right].$

Initial conditions: $y(0) = c_1 = 3; \dot{y}(0) = 3c_2 - 3 = 0 \implies c_2 = 1.$

**Solution:** $\mathbf{y(t) = e^{-t}\left(3\cos 3t + \sin 3t\right)}$

**(viii)** $2\ddot{z} + 7\dot{z} - 4z = 0$ **with** $z(0) = 0,\ \dot{z}(0) = 9$

Auxiliary eq: $2r^2 + 7r - 4 = 0 \implies (2r-1)(r+4) = 0 \implies r_1 = \frac{1}{2}, r_2 = -4.$

General solution: $z(t) = c_1 e^{t/2} + c_2 e^{-4t} \implies \dot{z}(t) = \frac{1}{2}c_1 e^{t/2} - 4c_2 e^{-4t}.$

Initial conditions: $z(0) = c_1 + c_2 = 0 \implies c_2 = -c_1; \dot{z}(0) = \frac{1}{2}c_1 + 4c_1 = \frac{9}{2}c_1 = 9 \implies c_1 = 2, c_2 = -2.$

**Solution:** $\mathbf{z(t) = 2e^{t/2} - 2e^{-4t}}$

**(ix)** $\ddot{y} + 2\dot{y} + y = 0$ **with** $y(0) = 0,\ \dot{y}(0) = -1$

Auxiliary eq: $r^2 + 2r + 1 = 0 \implies (r+1)^2 = 0 \implies r = -1$ (mult. 2).

General solution: $y(t) = (c_1 + c_2 t)e^{-t} \implies \dot{y}(t) = (c_2 - c_1 - c_2 t)e^{-t}.$

Initial conditions: $y(0) = c_1 = 0; \dot{y}(0) = c_2 - 0 = -1 \implies c_2 = -1.$

**Solution:** $\mathbf{y(t) = -te^{-t}}$

**(x)** $\ddot{x} + 6\dot{x} + 10x = 0$ **with** $x(0) = 3,\ \dot{x}(0) = 1$

Auxiliary eq: $r^2 + 6r + 10 = 0 \implies (r+3)^2 + 1 = 0 \implies r = -3 \pm i\ (\rho = -3, \omega = 1).$

General solution: $x(t) = e^{-3t}(c_1\cos t + c_2 \sin t).$

Derivative: $\dot{x}(t) = e^{-3t}\left[(c_2 - 3c_1)\cos t - (c_1 + 3c_2)\sin t\right].$

Initial conditions: $x(0) = c_1 = 3; \dot{x}(0) = c_2 - 3(3) = 1 \implies c_2 = 10.$

**Solution:** $\mathbf{x(t) = e^{-3t}\left(3\cos t + 10\sin t\right)}$

**(xi)** $4\ddot{x} - 20\dot{x} + 21x = 0$ **with** $x(0) = -4,\ \dot{x}(0) = -12$

Auxiliary eq: $4r^2 - 20r + 21 = 0. \Delta = 400 - 336 = 64 = 8^2 \implies r = \frac{20 \pm 8}{8} \implies r_1 = \frac{7}{2}, r_2 = \frac{3}{2}.$

General solution: $x(t) = c_1 e^{3t/2} + c_2 e^{7t/2} \implies \dot{x}(t) = \frac{3}{2}c_1 e^{3t/2} + \frac{7}{2}c_2 e^{7t/2}.$

Initial conditions: $c_1 + c_2 = -4, \frac{3}{2}c_1 + \frac{7}{2}(-4 - c_1) = -2c_1 - 14 = -12 \implies -2c_1 = 2 \implies c_1 = -1,\ c_2 = -3.$

**Solution:** $\mathbf{x(t) = -e^{3t/2} - 3e^{7t/2}}$

**(xii)** $\ddot{y} + \dot{y} - 2y = 0$ **with** $y(0) = 4,\ \dot{y}(0) = -4$

Auxiliary eq: $r^2 + r - 2 = 0 \implies (r+2)(r-1) = 0 \implies r_1 = 1, r_2 = -2.$

General solution: $y(t) = c_1 e^{t} + c_2 e^{-2t} \implies \dot{y}(t) = c_1 e^{t} - 2c_2 e^{-2t}.$

Initial conditions: $c_1 + c_2 = 4, c_1 - 2c_2 = -4 \implies 3c_2 = 8 \implies c_2 = \frac{8}{3},\ c_1 = \frac{4}{3}.$

**Solution:** $\mathbf{y(t) = \frac{4}{3}e^{t} + \frac{8}{3}e^{-2t}}$

**(xiii)** $\ddot{y} - 4y = 0$ **with** $y(0) = 10,\ \dot{y}(0) = 0$

Auxiliary eq: $r^2 - 4 = 0 \implies r = \pm 2.$

General solution: $y(t) = c_1 e^{2t} + c_2 e^{-2t} \implies \dot{y}(t) = 2c_1 e^{2t} - 2c_2 e^{-2t}.$

Initial conditions: $c_1 + c_2 = 10, 2(c_1 - c_2) = 0 \implies c_1 = c_2 = 5.$

**Solution:** $\mathbf{y(t) = 5e^{2t} + 5e^{-2t} = 10\cosh(2t)}$

**(xiv)** $y'' + 4y' + 4y = 0$ **with** $y(0) = 27,\ y'(0) = -54$

Auxiliary eq: $r^2 + 4r + 4 = 0 \implies (r+2)^2 = 0 \implies r = -2$ (mult. 2).

General solution: $y(t) = (c_1 + c_2 t)e^{-2t} \implies y'(t) = (c_2 - 2c_1 - 2c_2 t)e^{-2t}.$

Initial conditions: $y(0) = c_1 = 27; y'(0) = c_2 - 2(27) = -54 \implies c_2 = 0.$

**Solution:** $\mathbf{y(t) = 27e^{-2t}}$

**(xv)** $\ddot{y} + \omega^2 y = 0$ **with** $y(0) = 0,\ \dot{y}(0) = 1$ **(** $\omega \neq 0$ **)**

Auxiliary eq: $r^2 + \omega^2 = 0 \implies r = \pm i\omega.$

General solution: $y(t) = c_1\cos(\omega t) + c_2\sin(\omega t) \implies \dot{y}(t) = -c_1\omega\sin(\omega t) + c_2\omega\cos(\omega t).$

Initial conditions: $y(0) = c_1 = 0; \dot{y}(0) = c_2\omega = 1 \implies c_2 = \frac{1}{\omega}.$

**Solution:** $\mathbf{y(t) = \frac{1}{\omega}\sin(\omega t)}$

PHASE 4: MASTER SUMMARY TABLE

| Item | Differential Equation | Initial Conditions | Roots $r$ | Unique IVP Solution |
|---|---|---|---|---|
| (i) | $\ddot{x} - 3\dot{x} + 2x = 0$ | $x(0) = 2, \dot{x}(0) = 6$ | $1, 2$ | $\mathbf{-2e^{t} + 4e^{2t}}$ |
| (ii) | $y'' - 4y' + 4y = 0$ | $y(0) = 0, y'(0) = 3$ | $2$ (mult. 2) | $\mathbf{3te^{2t}}$ |
| (iii) | $z'' - 4z' + 13z = 0$ | $z(0) = 7, z'(0) = 42$ | $2 \pm 3i$ | $\mathbf{e^{2t}(7\cos 3t + \frac{28}{3}\sin 3t)}$ |
| (iv) | $\ddot{y} + \dot{y} - 6y = 0$ | $y(0) = -1, y'(0) = 8$ | $2, -3$ | $\mathbf{e^{2t} - 2e^{-3t}}$ |
| (v) | $\ddot{y} - 4\dot{y} = 0$ | $y(0) = 13, \dot{y}(0) = 0$ | $0, 4$ | $\mathbf{13}$ |
| (vi) | $\ddot{\theta} + 4\theta = 0$ | $\theta(0) = 0, \dot{\theta}(0) = 10$ | $\pm 2i$ | $\mathbf{5\sin 2t}$ |
| (vii) | $\ddot{y} + 2\dot{y} + 10y = 0$ | $y(0) = 3, \dot{y}(0) = 0$ | $-1 \pm 3i$ | $\mathbf{e^{-t}(3\cos 3t + \sin 3t)}$ |
| (viii) | $2\ddot{z} + 7\dot{z} - 4z = 0$ | $z(0) = 0, \dot{z}(0) = 9$ | $\frac{1}{2}, -4$ | $\mathbf{2e^{t/2} - 2e^{-4t}}$ |
| (ix) | $\ddot{y} + 2\dot{y} + y = 0$ | $y(0) = 0, \dot{y}(0) = -1$ | $-1$ (mult. 2) | $\mathbf{-te^{-t}}$ |
| (x) | $\ddot{x} + 6\dot{x} + 10x = 0$ | $x(0) = 3, \dot{x}(0) = 1$ | $-3 \pm i$ | $\mathbf{e^{-3t}(3\cos t + 10\sin t)}$ |
| (xi) | $4\ddot{x} - 20\dot{x} + 21x = 0$ | $x(0) = -4, \dot{x}(0) = -12$ | $\frac{3}{2}, \frac{7}{2}$ | $\mathbf{-e^{3t/2} - 3e^{7t/2}}$ |
| (xii) | $\ddot{y} + \dot{y} - 2y = 0$ | $y(0) = 4, \dot{y}(0) = -4$ | $1, -2$ | $\mathbf{\frac{4}{3}e^{t} + \frac{8}{3}e^{-2t}}$ |
| (xiii) | $\ddot{y} - 4y = 0$ | $y(0) = 10, \dot{y}(0) = 0$ | $\pm 2$ | $\mathbf{10\cosh(2t)}$ |
| (xiv) | $y'' + 4y' + 4y = 0$ | $y(0) = 27, y'(0) = -54$ | $-2$ (mult. 2) | $\mathbf{27e^{-2t}}$ |
| (xv) | $\ddot{y} + \omega^2 y = 0$ | $y(0) = 0, \dot{y}(0) = 1$ | $\pm i\omega$ | $\mathbf{\frac{1}{\omega}\sin(\omega t)}$ |

EXERCISE 5

Phase Plane • Stable Separatrix

# Saddle Point Invariant Manifold and Asymptotic Decay

Original statement · ProblemsCh3.pdf, p. 2

PHASE 1: STATEMENT & SETUP

> If the roots of the auxiliary equation are $k_1 > 0$ and $-k_2 < 0$ then the solution is $x(t) = Ae^{k_1 t} + Be^{-k_2 t}$. For most choices of initial conditions $x(0) = x_0,\ \dot{x}(0) = y_0$, we will have $x(t) \to \pm\infty$ as $t \to \infty$. However, there are some special initial conditions for which $x(t) \to 0$ as $t \to \infty$. Find the relationship between $x_0$ and $y_0$ that ensures this.

PHASE 2: FUNDAMENTAL FORMULATION

Given $k_1 > 0$ and $k_2 > 0$, the mode $e^{k_1 t}$ diverges exponentially to $+\infty$, while $e^{-k_2 t} \to 0$ as $t \to \infty$. Therefore, $x(t) \to 0$ if and only if the unstable amplitude is strictly annihilated: $A = 0$.

PHASE 3: STEP-BY-STEP ANALYTICAL DERIVATION

**Step 1: Differentiate the general solution:**

$$x(t) = Ae^{k_1 t} + Be^{-k_2 t} \implies \dot{x}(t) = k_1 Ae^{k_1 t} - k_2 Be^{-k_2 t}$$

**Step 2: Impose initial Cauchy conditions at $t = 0$:**

$$x(0) = A + B = x_0 \tag{1}$$

$$\dot{x}(0) = k_1 A - k_2 B = y_0 \tag{2}$$

**Step 3: Solve for the amplitude $A$:**

Multiply (1) by $k_2$ and add to (2):

$$k_2(A + B) + (k_1 A - k_2 B) = k_2 x_0 + y_0 \implies (k_1 + k_2)A = y_0 + k_2 x_0$$

$$A = \frac{y_0 + k_2 x_0}{k_1 + k_2} \tag{3}$$

**Step 4: Asymptotic Condition $x(t) \to 0$:**

Because $\lim_{t\to\infty} e^{k_1 t} = \infty$ and $\lim_{t\to\infty} e^{-k_2 t} = 0$:

$$\lim_{t\to\infty} x(t) = 0 \iff A = 0$$

$$\frac{y_0 + k_2 x_0}{k_1 + k_2} = 0 \implies \mathbf{y_0 = -k_2 x_0}$$

PHASE 4: INTERPRETATION & DYNAMICAL SYSTEMS

ASYMPTOTIC DECAY CONDITION

$$\boxed{y_0 = -k_2 x_0 \iff \dot{x}(0) = -k_2 x(0)}$$

When this relationship holds, $A = 0$ and $B = x_0$, giving the purely decaying trajectory $\mathbf{x(t) = x_0 e^{-k_2 t}}$.

GEOMETRIC PHASE PLANE MEANING: SADDLE SEPARATRIX

In the phase plane $(x, \dot{x})$, the origin is a saddle equilibrium. The line $\dot{x} = -k_2 x$ is the **stable invariant manifold** (separatrix). Any state starting off this 1D manifold is swept away by the unstable manifold $\dot{x} = k_1 x$, diverging exponentially to infinity.

EXERCISE 6

Polar Representation • Phasor Geometry

# Amplitude-Phase Transformation for Harmonic Oscillations

Original statement · ProblemsCh3.pdf, p. 2

PHASE 1: STATEMENT & SETUP

> Given an oscillating function of the form $x(t) = A\cos(\omega t) + B\sin(\omega t)$, show that it can be rewritten as a "single oscillation" $x(t) = M\cos(\omega t - \phi)$ with amplitude $M = (A^2 + B^2)^{1/2}$ and phase $\phi = \arctan(B/A)$.

PHASE 2: FUNDAMENTAL FORMULATION

Apply the angle subtraction identity $\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$ with $\alpha = \omega t$ and $\beta = \phi$, and equate coefficients with the Cartesian basis $\{\cos\omega t, \sin\omega t\}$.

PHASE 3: STEP-BY-STEP ANALYTICAL DERIVATION

**Step 1: Expand the single oscillation form:**

$$x(t) = M\cos(\omega t - \phi) = M\left[\cos(\omega t)\cos\phi + \sin(\omega t)\sin\phi\right] = (M\cos\phi)\cos(\omega t) + (M\sin\phi)\sin(\omega t)$$

**Step 2: Equate with Cartesian components:**

$$A\cos(\omega t) + B\sin(\omega t) = (M\cos\phi)\cos(\omega t) + (M\sin\phi)\sin(\omega t)$$

$$\begin{cases} A = M\cos\phi & (1) \\ B = M\sin\phi & (2) \end{cases}$$

**Step 3: Solve for Amplitude $M$:**

Square and add (1) and (2):

$$A^2 + B^2 = M^2\cos^2\phi + M^2\sin^2\phi = M^2(\cos^2\phi + \sin^2\phi) = M^2$$

$$\mathbf{M} = \quad \mathbf{A^2 + B^2} = (\mathbf{A^2 + B^2})^{\mathbf{1/2}}$$

**Step 4: Solve for Phase Lag $\phi$:**

Divide (2) by (1):

$$\frac{B}{A} = \frac{M \sin\phi}{M \cos\phi} = \tan\phi \implies \phi = \arctan\left(\frac{\mathbf{B}}{\mathbf{A}}\right)$$

PHASE 4: INTERPRETATION & ENGINEERING RELEVANCE

**POLAR HARMONIC FORMULAS**

$$\boxed{x(t) = M\cos(\omega t - \phi), \qquad M = \quad A^2 + B^2, \qquad \phi = \arctan(B/A)}$$

**STRUCTURAL DYNAMICS APPLICATION**

In aircraft wing flutter and structural modal analysis, the amplitude-phase form immediately reveals the maximum physical peak deflection $x_{\max} = M$ and time lag $\Delta t = \phi/\omega$ relative to external periodic gusts.

EXERCISE 7

Energy First Integral • Pendulum Isochronism

# Simple Pendulum Energy Conservation and Small Oscillations

Original statement · ProblemsCh3.pdf, p. 2

PHASE 1: STATEMENT & SETUP

Consider a so-called simple pendulum whose shaft has length $L$ and negligible mass, and a suspended mass $m$ at its end. If the pivot is at the origin of coordinates, show that:

1. The gravitational potential energy is $V(\theta) = mgL(1 - \cos\theta)$, where $\theta$ is the angle with the downward vertical.
2. The total energy is $E = \frac{1}{2}mL^2\dot{\theta}^2 + mgL(1 - \cos\theta)$.
3. The equation of motion reads $\frac{d^2\theta}{dt^2} = -\frac{g}{L}\sin\theta$, obtained by requesting $\frac{dE}{dt} = 0$.
4. By considering small oscillations around the vertical, the equation becomes $\frac{d^2\theta}{dt^2} = -\frac{g}{L}\theta$. Show that the general solution reads $\theta(t) = M\cos(\omega t - \phi)$ and relate $\omega$ with system parameters. Does $\omega$ depend on mass $m$? Does it depend on amplitude?

PHASE 2: FUNDAMENTAL FORMULATION

Let pivot be at $(0, 0)$ and positive $y$ upwards. Position of bob: $x = L\sin\theta,\ y = -L\cos\theta$. Velocity along arc: $v = L\dot{\theta}$. Total mechanical energy $E = T + V$. In the absence of dissipative friction, mechanical energy is conserved: $\frac{dE}{dt} = 0$.

PHASE 3: STEP-BY-STEP ANALYTICAL DERIVATION

**Part (i): Potential Energy $V(\theta)$**

Vertical height above the lowest point $\theta = 0$ ($y_{\min} = -L$):

$$h(\theta) = y(\theta) - (-L) = -L\cos\theta + L = L(1 - \cos\theta)$$

$$\mathbf{V(\theta) = mgh = mgL(1 - \cos\theta)}$$

**Part (ii): Total Energy $E$**

Kinetic energy of point mass: $T = \frac{1}{2}mv^2 = \frac{1}{2}m(L\dot{\theta})^2 = \frac{1}{2}mL^2\dot{\theta}^2$.

$$\mathbf{E = T + V(\theta) = \frac{1}{2}mL^2\dot{\theta}^2 + mgL(1 - \cos\theta)}$$

**Part (iii): Equation of Motion via Energy Conservation**

Differentiating $E(t)$ with respect to time using the chain rule:

$$\frac{dE}{dt} = \frac{1}{2}mL^2(2\dot{\theta}\ddot{\theta}) + mgL(\sin\theta \cdot \dot{\theta}) = mL^2\dot{\theta}\ddot{\theta} + mgL\dot{\theta}\sin\theta = 0$$

Factoring out $mL^2\dot{\theta} \neq 0$:

$$mL^2\dot{\theta}\left(\ddot{\theta} + \frac{g}{L}\sin\theta\right) = 0 \implies \mathbf{\frac{d^2\theta}{dt^2} = -\frac{g}{L}\sin\theta}$$

**Part (iv): Small Oscillations Linearization & Frequency Analysis**

For $|\theta| \ll 1$, $\sin\theta \approx \theta$. The ODE simplifies to:

$$\ddot{\theta} + \frac{g}{L}\theta = 0 \iff \ddot{\theta} + \omega^2\theta = 0, \quad \text{where } \omega = \mathbf{\sqrt{\frac{g}{L}}}$$

The general solution is $\theta(t) = c_1\cos\omega t + c_2\sin\omega t$. In amplitude-phase form:

$$\mathbf{\theta(t) = M\cos(\omega t - \phi)}$$

PHASE 4: CONCEPTUAL ANSWERS & MASTER RESULTS

NATURAL FREQUENCY & ANSWERS TO QUESTIONS

$$\omega = \sqrt{\frac{g}{L}} \quad [\text{rad/s}], \qquad T_0 = 2\pi \sqrt{\frac{L}{g}} \quad [\text{s}]$$

**1. Does frequency depend on mass $m$? NO.** Mass $m$ cancels identically from both terms in the equation of motion.

**2. Does frequency depend on amplitude $M$?** In the *linear small-amplitude approximation*, **NO** (Galileo's isochronism). In the *exact nonlinear pendulum*, the period lengthens with amplitude according to complete elliptic integrals: $T(M) \approx T_0(1 + \frac{1}{16}M^2 + \dots)$.

EXERCISE 8

Mechanical Damping • Under/Critical/Overdamped

# Physical Regimes of the Damped Harmonic Oscillator

Original statement · ProblemsCh3.pdf, p. 2

PHASE 1: STATEMENT & SETUP

Consider the equation $m\ddot{x} + \mu\dot{x} + kx = 0$ for the deviation $x(t)$ of a point particle with mass $m$ from its equilibrium position, subject to an elastic force (constant $k$) and drag force (parameter $\mu$). Discuss how the form of the solutions changes depending on the value of $\mu$ relative to the critical damping $\mu_c = 2 \quad mk$.

PHASE 2: FUNDAMENTAL FORMULATION

The auxiliary polynomial is $mr^2 + \mu r + k = 0$. The discriminant is $\Delta = \mu^2 - 4mk = \mu^2 - \mu_c^2$, where $\mu_c = 2 \quad mk$. Roots are $r_{1,2} = -\frac{\mu}{2m} \pm \frac{\mu^2 - \mu_c^2}{2m} = -\gamma \pm \quad \gamma^2 - \omega_0^2$.

PHASE 3: STEP-BY-STEP ANALYTICAL DERIVATION

**1. Underdamped Regime:** $\mu < \mu_c = 2 \quad mk$ **(** $\Delta < 0$ **)**

The discriminant is negative: $\quad \mu^2 - 4mk = i \quad 4mk - \mu^2$.

Roots are complex conjugates with negative real part: $r_{1,2} = -\gamma \pm i\omega_d$, where:

$$\gamma = \frac{\mu}{2m}, \qquad \omega_d = \frac{4mk - \mu^2}{2m} = \quad \omega_0^2 - \gamma^2$$

General solution:

$$\mathbf{x(t) = e^{-\gamma t}\left(c_1 \cos \omega_d t + c_2 \sin \omega_d t\right) = Me^{-\gamma t} \cos(\omega_d t - \phi)}$$

*Behavior:* Decaying oscillatory motion within an exponential envelope $\pm Me^{-\gamma t}$, crossing zero infinitely many times.

**2. Critically Damped Regime:** $\mu = \mu_c = 2 \quad mk$ **(** $\Delta = 0$ **)**

The discriminant vanishes identically. The characteristic equation has a single repeated real root:

$$r_1 = r_2 = -\frac{\mu_c}{2m} = - \quad \frac{k}{m} = -\omega_0$$

General solution:

$$\mathbf{x(t) = (c_1 + c_2 t)e^{-\omega_0 t}}$$

*Behavior:* Non-oscillatory. Crosses equilibrium at most once. Represents the fastest monotonic return to equilibrium without overshoot.

### 3. Overdamped Regime: $\mu > \mu_c = 2 \quad mk$ ( $\Delta > 0$ )

The discriminant is positive: $\quad \mu^2 - 4mk < \mu$. Both roots are distinct, real, and strictly negative:

$$r_1 = -\gamma + \quad \gamma^2 - \omega_0^2 < 0, \qquad r_2 = -\gamma - \quad \gamma^2 - \omega_0^2 < 0$$

General solution:

$$\mathbf{x(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}}$$

*Behavior:* Purely exponential sluggish decay. At long times, dominated by the slower decay mode $e^{r_1 t}$. Crosses zero at most once.

PHASE 4: MASTER SUMMARY & ENGINEERING IMPACT

| Damping Regime | Damping Condition | Roots $r_{1,2}$ | General Solution $x(t)$ | Dynamic Behavior |
|---|---|---|---|---|
| **Underdamped** | $\mu < 2 \quad mk$ | $-\gamma \pm i\omega_d$ | $Me^{-\gamma t}\cos(\omega_d t - \phi)$ | Oscillatory decay, infinite crossings |
| **Critically Damped** | $\mu = 2 \quad mk$ | $-\omega_0$ (mult. 2) | $(c_1 + c_2 t)e^{-\omega_0 t}$ | Fastest return to trim without overshoot |
| **Overdamped** | $\mu > 2 \quad mk$ | $r_2 < r_1 < 0$ | $c_1 e^{r_1 t} + c_2 e^{r_2 t}$ | Sluggish exponential relaxation, no oscillations |

EXERCISE 9

Hyperbolic Motion • Variable Hanging Mass

# Falling Chain from Table and Hyperbolic Motion

Original statement · ProblemsCh3.pdf, p. 3

PHASE 1: STATEMENT & SETUP

A chain of length $\ell = 5$ m and mass $m$, with uniform mass density, is sitting on a table. If the chain is initially at rest and $1$ m of the chain hangs from the edge of the table, apply Newton's second law to calculate the time $t_*$ when the whole chain falls off the table, neglecting any friction.

*Hint:* Formulate a second order ODE for the variable $x(t)$, which is the length of the portion of chain that is overhanging, as a function of time.

PHASE 2: FUNDAMENTAL FORMULATION

Let $x(t) \in [1, 5]$ be the overhanging length. Mass density is $\lambda = m/\ell$. The downward gravity force pulling the chain is $F_{\text{ext}} = m_{\text{hang}} g = (\lambda x) g = \frac{m}{\ell} g x$. Because the table is frictionless and the entire chain of constant total mass $m$ accelerates together along its contour:

$$m\ddot{x} = \frac{m}{\ell} g x \implies \ddot{x} - \frac{g}{\ell} x = 0$$

PHASE 3: STEP-BY-STEP ANALYTICAL DERIVATION

**Step 1: Auxiliary Equation and General Solution**

Let $\alpha = \sqrt{\frac{g}{\ell}}$. The ODE is $\ddot{x} - \alpha^2 x = 0$.

Auxiliary eq: $r^2 - \alpha^2 = 0 \implies r = \pm\alpha$.

Hyperbolic general solution:

$$x(t) = c_1 \cosh(\alpha t) + c_2 \sinh(\alpha t)$$

$$\dot{x}(t) = \alpha\left[c_1 \sinh(\alpha t) + c_2 \cosh(\alpha t)\right]$$

**Step 2: Initial Conditions**

Released from rest: $\dot{x}(0) = \alpha c_2 = 0 \implies c_2 = 0$.

Initial hanging length: $x(0) = c_1 = x_0 = 1\ \text{m}$.

$$\mathbf{x(t) = x_0 \cosh(\alpha t) = \cosh\left(\quad \frac{g}{\ell}\, t\right)}$$

**Step 3: Detachment Time $t_*$**

The entire chain falls off when $x(t_*) = \ell = 5\ \text{m}$:

$$\cosh(\alpha t_*) = \frac{\ell}{x_0} = \frac{5}{1} = 5$$

$$\alpha t_* = \operatorname{arcosh}(5) = \ln\left(5 + \quad 5^2 - 1\right) = \ln(5 + \quad 24) = \ln(5 + 2 \quad 6)$$

**Step 4: Numerical Evaluation**

Using standard gravity $g = 9.8\ \text{m/s}^2$ and $\ell = 5\ \text{m}$:

$$\alpha = \quad \frac{9.8}{5} = \quad 1.96 = 1.4\ \text{s}^{-1}$$

$$\ln(5 + 2 \quad 6) \approx \ln(5 + 4.898979) = \ln(9.898979) \approx 2.292432$$

$$\mathbf{t_* = \frac{2.292432}{1.4} \approx 1.63745\ \text{s} \approx 1.637\ \text{s}}$$

*(If taking $g = 9.81\ \text{m/s}^2$, $\alpha \approx 1.40071\ \text{s}^{-1} \implies t_* \approx 1.6366\ \text{s} \approx 1.637\ \text{s}$)*

PHASE 4: INTERPRETATION & CONSERVATION CHECK

MASTER FALL TIME & DETACHMENT VELOCITY

$$\boxed{t_* = \frac{\operatorname{arcosh}(5)}{1.4} \approx 1.637 \text{ s}}$$

**Detachment velocity:** $v_* = \dot{x}(t_*) = x_0 \alpha \sinh(\alpha t_*) = 1 \times 1.4 \times \quad 5^2 - 1 = 1.4 \quad 24 \approx \mathbf{6.859}$ m/s.

**Energy Check:** $\Delta E_p = mg\frac{\ell}{2} - mg\frac{x_0^2}{2\ell} = \frac{mg}{2\ell}(\ell^2 - x_0^2) = \frac{1}{2}mv_*^2 \implies v_* = \quad \frac{g}{\ell}(\ell^2 - x_0^2) = 1.96(24) \approx 6.859$ m/s. Exact match!

EXERCISE 10

Linear Drag • Terminal Velocity

# Parachutist with Linear Aerodynamic Drag and Terminal Speed

Original statement · ProblemsCh3.pdf, p. 3

PHASE 1: STATEMENT & SETUP

> A parachutist of mass $m$ kg jumps from a plane with initial velocity $0$. Because of the parachute, there is a friction force that avoids free fall; this friction turns out to be proportional (with constant of proportionality $c > 0$) to the velocity. Write down an ODE to describe the motion of the parachutist, and determine the terminal speed (when $t \to \infty$). Compare this speed with that of free fall.

PHASE 2: FUNDAMENTAL FORMULATION

Taking positive axis downwards, Newton's 2nd law gives:

$$\sum F = m\frac{dv}{dt} = mg - cv \implies \frac{dv}{dt} + \frac{c}{m}v = g$$

With initial condition $v(0) = 0$.

PHASE 3: STEP-BY-STEP ANALYTICAL DERIVATION

**Step 1: Integrating Factor Method**

Integrating factor $\mu(t) = e^{(c/m)t}$:

$$\frac{d}{dt}\left[v(t)e^{(c/m)t}\right] = ge^{(c/m)t}$$

$$v(t)e^{(c/m)t} = \int ge^{(c/m)t}\,dt = \frac{mg}{c}e^{(c/m)t} + C$$

$$v(t) = \frac{mg}{c} + Ce^{-(c/m)t}$$

**Step 2: Initial Condition $v(0) = 0$**

$$v(0) = \frac{mg}{c} + C = 0 \implies C = -\frac{mg}{c}$$

$$\mathbf{v(t) = \frac{mg}{c}\left(1 - e^{-(c/m)t}\right)}$$

**Step 3: Terminal Velocity ($t \to \infty$)**

As $t \to \infty$, $e^{-(c/m)t} \to 0$:

$$\mathbf{v_\infty = \lim_{t\to\infty} v(t) = \frac{mg}{c}}$$

**Step 4: Comparison with Free Fall**

- **Free fall ($c = 0$):** $v_{\text{free}}(t) = gt$. Velocity increases linearly without bound ($v \to \infty$).
- **Parachutist with Drag ($c > 0$):** Velocity saturates asymptotically to a safe finite constant $v_\infty = \frac{mg}{c}$.
- **Short-time Taylor expansion ($t \ll m/c$):**

$$v(t) = \frac{mg}{c}\left[1 - \left(1 - \frac{c}{m}t + \frac{c^2}{2m^2}t^2 - \dots\right)\right] = gt - \frac{cg}{2m}t^2 + \dots$$

At the first instant ($t \to 0$), the jumper accelerates at $g$, identically to free fall, before drag builds up.

PHASE 4: MASTER SUMMARY & FLIGHT MECHANICS

**TERMINAL VELOCITY & VELOCITY PROFILE**

$$\boxed{\frac{dv}{dt} + \frac{c}{m}v = g \implies v(t) = \frac{mg}{c}\left(1 - e^{-(c/m)t}\right)}$$

$$\boxed{v_\infty = \frac{mg}{c}}$$

EXERCISE 11

Euler-Cauchy • Coordinate Mapping • Resonance

# Euler-Cauchy Equation via Exponential Transformation

Original statement · ProblemsCh3.pdf, p. 3

PHASE 1: STATEMENT & SETUP

We consider the following second order ODE (Euler equation):

$$x^2y''(x) + pxy'(x) + qy(x) = g(x), \quad x > 0$$

1. By making the change of variable $x = e^z$ and using the chain rule, show that the ODE is transformed into one with constant coefficients in the variable $z$:

$$\frac{d^2y}{dz^2} + (p-1)\frac{dy}{dz} + qy = g(e^z)$$

2. Take $p = 3,\ q = -3$ and $g(x) = 4x$, make the previous transformation and compute the general solution.

**ERRATUM IN THE PRINTED SHEET**

In ProblemsCh3.pdf, Exercise 11 (i), the transformed equation is printed as $\frac{d^2y}{dz^2} + (p-1)\frac{dy}{dz} + y = g(e^z)$, i.e. with "$+\,y$". The correct coefficient is $q$ (the term $q\,y(x)$ of the original equation is unchanged by $x = e^z$), so the equation above, with "$+\,q\,y$", is the one derived in Phase 3 and used in part 2.

PHASE 2: FUNDAMENTAL FORMULATION

Let $x = e^z \iff z = \ln x$ for $x > 0$. The derivative with respect to $x$ is $\frac{dz}{dx} = \frac{1}{x}$. By applying the chain and product rules to $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$, the power pre-factors $x$ and $x^2$ cancel identically, transforming the variable-coefficient ODE into a linear ODE with constant coefficients.

PHASE 3: STEP-BY-STEP ANALYTICAL DERIVATION

## Part (i): Chain Rule Derivation of the Transformation

**First Derivative:**

$$\frac{dy}{dx} = \frac{dy}{dz}\frac{dz}{dx} = \frac{1}{x}\frac{dy}{dz} \implies x\frac{dy}{dx} = \frac{dy}{dz} \tag{1}$$

**Second Derivative:**

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{1}{x}\frac{dy}{dz}\right) = -\frac{1}{x^2}\frac{dy}{dz} + \frac{1}{x}\frac{d}{dx}\left(\frac{dy}{dz}\right) = -\frac{1}{x^2}\frac{dy}{dz} + \frac{1}{x}\left(\frac{d^2y}{dz^2}\frac{dz}{dx}\right)$$

$$\frac{d^2y}{dx^2} = -\frac{1}{x^2}\frac{dy}{dz} + \frac{1}{x^2}\frac{d^2y}{dz^2} = \frac{1}{x^2}\left(\frac{d^2y}{dz^2} - \frac{dy}{dz}\right) \implies x^2\frac{d^2y}{dx^2} = \frac{d^2y}{dz^2} - \frac{dy}{dz} \tag{2}$$

**Substitution into the Euler-Cauchy Equation:**

$$x^2y'' + pxy' + qy = \left(\frac{d^2y}{dz^2} - \frac{dy}{dz}\right) + p\left(\frac{dy}{dz}\right) + qy = g(e^z)$$

$$\mathbf{\frac{d^2y}{dz^2} + (p-1)\frac{dy}{dz} + qy = g(e^z)}$$

## Part (ii): Specific Case $p = 3,\ q = -3,\ g(x) = 4x$

**1. Transformed ODE in $z$:**

With $p - 1 = 3 - 1 = 2, q = -3$, and $g(e^z) = 4e^z$:

$$\frac{d^2y}{dz^2} + 2\frac{dy}{dz} - 3y = 4e^z$$

**2. Homogeneous Solution $y_h(z)$:**

Auxiliary eq: $r^2 + 2r - 3 = 0 \implies (r+3)(r-1) = 0 \implies r_1 = 1, r_2 = -3.$

$$y_h(z) = c_1e^z + c_2e^{-3z}$$

**3. Resonant Particular Solution $y_p(z)$:**

Since the forcing exponent $e^{1\cdot z}$ matches the characteristic root $r_1 = 1$, we test the resonant ansatz $y_p(z) = Ate^z$:

$$y_p'(z) = A(1+z)e^z, \qquad y_p''(z) = A(2+z)e^z$$

$$y_p'' + 2y_p' - 3y_p = Ae^z\left[(2+z) + 2(1+z) - 3z\right] = Ae^z\left[4 + 3z - 3z\right] = 4Ae^z$$

Equating to $4e^z$: $4A = 4 \implies A = 1 \implies \mathbf{y_p(z) = ze^z}$.

**4. General Solution in $z$:**

$$y(z) = c_1e^z + c_2e^{-3z} + ze^z$$

**5. Back-transformation to $x$ ($z = \ln x,\ e^z = x,\ e^{-3z} = x^{-3}$):**

$$\mathbf{y(x) = c_1x + \frac{c_2}{x^3} + x\ln x, \quad x > 0}$$

PHASE 4: MASTER VERIFICATION & AEROSPACE APPLICATION

MASTER SOLUTION & ODE VERIFICATION

$$\boxed{y(x) = c_1x + c_2x^{-3} + x\ln x, \quad x > 0}$$

**Direct Verification:**

$y'(x) = c_1 - 3c_2x^{-4} + \ln x + 1$

$y''(x) = 12c_2x^{-5} + \frac{1}{x}$

$x^2y'' + 3xy' - 3y = (12c_2x^{-3} + x) + (3c_1x - 9c_2x^{-3} + 3x\ln x + 3x) - (3c_1x + 3c_2x^{-3} + 3x\ln x) = 4x \equiv g(x)$. Confirmed!

← PREVIOUS PROBLEM SET
Chapter 2: First-Order ODEs

REVIEW FOUNDATIONS →
Topic 3 Theory & Lecture Guide

Study notes written from the official course material (ProblemsCh3.pdf; Robinson (2004).pdf). Online version: subjects/advanced-maths/problemas/topic-3-problems.html